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Assignment

Deepfake Technology as a Cybersecurity Threat

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Abstract

Deepfake technology is something that uses artificial intelligence and deep learning, and it has grown very fast from a small research idea into a big cybersecurity problem. It uses things like GANs (Generative Adversarial Networks) and other computer models to make very real looking fake videos, fake voice recordings, and even copied faces. Earlier this was only used inside research labs, but now it is everywhere in internet. Because of this, some people are misusing it for identity fraud, phishing scams, company spying, and also spreading fake news on large scale. Now even people with very less technical knowledge can use these tools and create fake content that can trick both normal people and security systems. So now the main problem is more about human trust rather than only computer systems.

This report is about deepfake technology as a growing cybersecurity problem. It explains how it started in 2014 and how it has become more advanced in 2025. It also talks about different types of attacks using deepfakes like fake business emails, voice cloning scams, and fake identity creation. The report also covers how people are trying to detect deepfakes today, what kind of laws different countries are bringing, and what may happen in future. At the end, it says we need technology, rules, and awareness from people together to reduce the problems caused by deepfakes.

1. Introduction

Artificial intelligence has grown a lot in recent years. Because of that, many powerful technologies have been created. But at the same time, it has also brought some problems for cybersecurity. One of the most important examples is deepfake technology. The word “deepfake” comes from ‘deep learning’ and ‘fake’. It is used to describe AI tools that can create or change media content so that it looks real to people.

At the beginning, deepfakes were mainly used in research. Most of the work was done in areas related to generative models. But later, the technology started improving very fast. This was mainly because computers became more powerful, large datasets became available, and many machine learning tools were released as open-source [1]. Because of this, deepfake tools became easier for people to use.

In cybersecurity, deepfakes are considered a new type of threat. Traditional cyberattacks usually target systems, like software bugs or network problems. But deepfake attacks are different. They focus more on people. People usually trust what they see and hear, and deepfakes take advantage of that.

Because of this reason, deepfakes are risky in social engineering attacks. In these attacks, the goal is to influence people, not directly attack a system [2]. The results can be serious. It can cause financial loss, especially when attackers pretend to be company executives in business email compromise (BEC) scams. It can also lead to reputational damage, political problems, and loss of trust in digital media [3].

In recent years, deepfake technology has become more important in cybersecurity. A report by Deeptrace Labs showed that the number of deepfake videos online increased by more than 900% between 2018 and 2020. After that, the number has continued to grow, and the quality has also improved [4].

There are real examples that show how serious this issue is. For example, AI voice cloning was used to scam a UK energy company of around £200,000. In some cases, deepfake video calls were used in corporate fraud involving large companies [5]. These examples show that deepfake cybercrime is already happening and is not just a theory.

Another problem is the imbalance in this situation. Creating strong security systems needs time, money, and skilled people. But creating deepfakes has become much easier. Even people with limited resources can do it now.

This creates a weakness in cybersecurity. It cannot be solved only with technical solutions. It needs support from different areas like technology, laws, education, and cooperation between countries.

This report explains deepfake technology as a cybersecurity threat. Section 2 looks at the technical basics, Section 3 explains how it has developed, Section 4 discusses future changes and possible solutions, and Section 5 gives the conclusion with some recommendations.

2. Evolution of Deepfake Technology as a Cybersecurity Threat

The growth of deepfake technology as a cybersecurity problem can be understood in different stages. Each stage shows improvements in AI technology, how easily people can use deepfake tools, and how the risks also increased over time. In this section, we look at how deepfake technology started in academic research and how it slowly developed into a serious threat situation by 2025.

Table 1: Evolution of Deepfake Technology (2014–2025)

Phase	Period	Key Development	Cybersecurity Impact
Academic Origins	2014–2017	GAN architecture introduced by Goodfellow	Minimal - confined to research labs
Consumer Accessibility	2017–2019	DeepFaceLab, FaceSwap released publicly	NCII attacks, blackmail, harassment
Corporate Fraud	2019–2021	Voice cloning, BEC scams emerged	£200k UK energy firm fraud, \$1.8B BEC losses
Multimodal Deepfakes	2021–2023	Diffusion models + LLMs integrated	\$25M Hong Kong corporate fraud
Contemporary Threat	2023–2025	Text-to-video (Sora), DaaS platforms	KYC bypass, election interference, state-sponsored ops



Figure 1: Growth in Deepfake-related Research Papers (2018–2025), showing dominance of deceptive applications

2.1 Origins: Generative Adversarial Networks and Academic Research (2014– 2017)

The idea behind deepfake technology started with Ian Goodfellow and his team in 2014 when they introduced Generative Adversarial Networks (GANs). In this method there are two parts, generator and discriminator. Generator tries to create fake data and discriminator checks whether it is real or fake. Both keep competing during training and slowly the generator becomes better and produces very realistic outputs. So basically, this became the base for deepfake systems later.

From around 2014 to 2017, most deepfake work was only in universities and research labs. At that time, systems could only make simple fake images and quality was not that strong to fully trick people. During this stage, methods like Variational Autoencoder (VAE) and conditional GAN were introduced, which improved image generation and control. But these needed high computing power and technical knowledge, so normal users could not use them easily. So risk was still not very serious at that time.

At that time, researchers were mainly focused on improving how models work, not thinking much about misuse. They used metrics like FID and Inception Score to check image quality. But social impact was not really considered much. Later only it became clear that once this technology leaves labs, it can be misused in many ways.

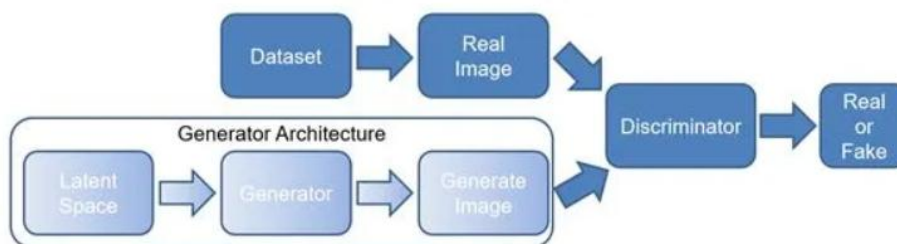


Figure 2: Generative Adversarial Network (GAN) Architecture — Generator and Discriminator components

2.2 Emergence of Consumer-Accessible Deepfakes (2017–2019)

So a big change in deepfake development happened in late 2017. At that time, an unknown user on Reddit called “deepfakes” shared some face swap videos made using AI. These videos were made using a simple method called autoencoder neural networks, and it was trained using images from the internet. After this, tools like DeepFaceLab and FaceSwap also came out as open-source. Because of that, anyone could easily try deepfakes, and this is where cybersecurity problems started becoming serious.

After that, it became much easier for normal people to create deepfake videos. Earlier it needed technical skills, but now even basic computer users could do it using ready-made software. In the beginning, most deepfakes online were used for wrong purposes like non-consensual content. Studies around 2019 showed that most of them (around 96%) were of this type, usually targeting private individuals, especially women. This clearly showed that deepfakes can be used for harassment, blackmail, and damaging someone’s reputation.

Later, mobile apps made it even more popular. Apps like FaceApp and Reface allowed people to swap faces very easily. Because of this, many users started thinking edited or fake media is just normal and fun. But the issue is, this also makes people less careful about what is real and what is fake, which is risky from a security point of view.

During this time, researchers also started focusing more on deepfakes. Some studies, like Tolosana et al. (2020), showed that deepfake creation was improving faster than detection methods. Simply put, fake content was getting better faster than systems trying to detect it. Even today, this gap is still a big cybersecurity problem.

2.3 Corporate and Financial Fraud Applications (2019–2021)

Between 2019 and 2021, deepfakes started being used in real cyberattacks mainly for money. One early case happened in 2019, where staff in a UK energy company got a phone call. The voice sounded exactly like their German parent company CEO. But later they found out it was actually an AI-generated voice. The attackers tricked them into sending around €220,000 to a bank account in Hungary. This showed how realistic voice cloning had become, even enough to fool experienced people during normal work calls.

Around the same time, deepfake videos also started being used in business scams. Attackers used fake videos to act like company executives in online meetings. In some cases, they told finance staff to approve fake money transfers. The FBI also reported that Business Email Compromise attacks increased a lot during this period, and losses went over \$1.8 billion in 2020, with some cases linked to deepfake use. So cybercrime was clearly becoming more advanced by mixing AI with social engineering tricks.

The main reason these attacks work is simple. People naturally trust what they see and hear. A familiar face or voice feels real, especially in work situations. Deepfakes take advantage of this by copying voices and faces very realistically, so people don't stop and think before reacting. Also in companies, employees sometimes feel pressure when instructions come from higher management, so they don't question it properly. This makes it easier for attackers.

During this time, deepfakes also started being used for spreading political fake news. Fake videos of leaders were shared on social media in different countries, which caused confusion and concern. Because of this, governments started treating deepfakes as a serious security threat. Some countries also began introducing laws to control this kind of misuse.

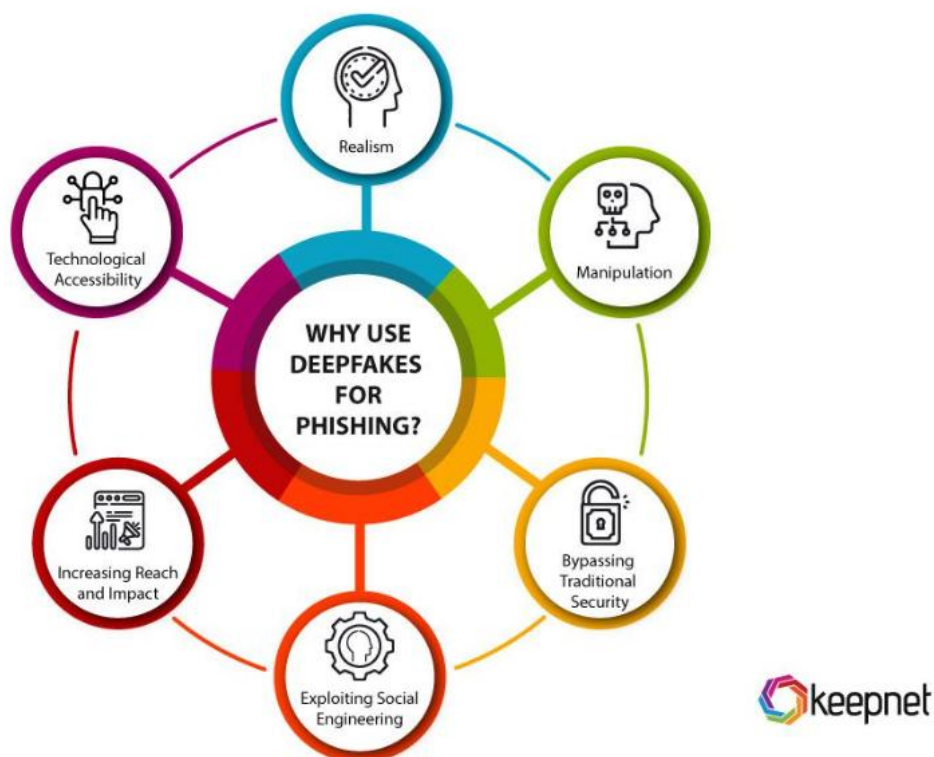


Figure 3: Key Reasons Why Deepfakes are Exploited in Phishing and BEC Attacks

Table 2: Key Deepfake Attack Vectors and Real-World Cases

Attack Type	Technique Used	Notable Case	Financial/Social Impact
Voice Cloning Fraud	AI voice synthesis	UK Energy Firm (2019)	€220,000 lost
BEC Video Scam	Multimodal deepfake video call	Hong Kong Company (2024)	\$25 million lost
KYC/Biometric Bypass	Synthetic facial imagery	Multiple financial institutions	Regulatory alerts issued
Political Disinformation	Synthetic video of politicians	Multiple elections globally	Public trust erosion
NCII Attacks	Face-swap algorithms	Widespread (2019)	96% of deepfakes online

2.4 Advanced Multimodal Deepfakes and Scaled Cybercrime (2021–2023)

Between 2021 and 2023, deepfake technology improved a lot. Earlier it was mostly just face or voice changes, but now it became more advanced where multiple things could be changed at the same time like face, voice, body movement, and even background. Around this time, new methods called diffusion models became popular. One example is Stable Diffusion, which gave better results compared to older GAN methods. Also, AI language models started combining with image and video tools, so now deepfakes could create more “story-like” and realistic content depending on the situation.

A serious case was reported in 2024 where a company in Hong Kong lost around \$25 million. An employee joined a video call thinking it was real, but actually everyone in the meeting including the CFO was fake AI-generated people. Because everything looked and sounded real, the employee ended up sending multiple bank transfers without realizing it was a scam. This showed how deepfakes had become so advanced that even full fake meetings can be created, not just single fake faces or voices.

At the same time, deepfake tools also became easier to access. Now there are services where people can just pay and get deepfake videos made without knowing any technical skills. These are called Deepfake-as-a-Service platforms. Some of them are even available on dark web marketplaces. The cost is also quite low, sometimes just a few hundred dollars, so even small criminals can use them. This made cybercrime easier and more common compared to earlier times when it needed high technical knowledge.

Detection methods also improved during this time. Researchers started looking for small signs like unusual blinking, strange eye movement, or unnatural facial expressions to detect fake videos. AI-based detection tools also became better and more accurate. But the problem is, new deepfake methods keep coming, so detection tools sometimes struggle with new types of fake content. So it becomes like a continuous race between creating deepfakes and detecting them.

2.5 Contemporary Threat Landscape (2023–2025)

From 2023 to 2025, deepfakes became a big problem in cybersecurity. Now anyone can make fake videos just by typing a few words. These videos look very real, so people easily get confused.

Deepfakes are also used with other tricks like fake emails, fake ID cards, and fake social media accounts. When all these are used together, the attack becomes more serious. It is hard for normal people to find out what is real.

In banks and online systems, deepfakes are causing issues. Some people have used fake faces to cheat face recognition systems. They try to open accounts or get access using this. Because of that, companies added more checks.

Deepfakes are also used in news and politics. Fake videos and fake voices can spread wrong information. Many people believe it without checking, and it creates confusion.

Now companies are more careful. Employees are told to double-check any unusual request, especially money transfers. They confirm it using another way like a phone call or message. Still, attackers keep improving, so companies are trying to stay alert.

3. Future Developments in Deepfake Cybersecurity

In future, deepfake technology will keep growing. As it improves, it will bring more problems in cybersecurity.

Fake videos will become more normal, and it will be harder to know what is real and what is not. People can easily get confused because everything looks real.

Because of this, online attacks may also increase and become more serious over time. So companies will have to be more careful. They need better ways to catch fake videos and protect users from these kinds of problems.

Table 3: Future Developments in Deepfake Cybersecurity

Development	Expected Timeline	Threat/Opportunity	Key Research
Real-Time Mobile Deepfakes	2025–2027	High Threat - live call fraud	Edge computing acceleration
C2PA Media Provenance	2025–2026	Defensive Opportunity	Cryptographic content signing
EU AI Act Enforcement	2025 onwards	Regulatory Milestone	Transparency labelling required
AI Detection at Scale	2026–2028	Defensive Opportunity	Cross-method generalization
Quantum-Accelerated Generation	2030+	Critical Future Threat	Quantum ML research

3.1 Advancement of Real-Time Deepfake Generation

One important future change in deepfakes is real-time generation. This means fake videos and voices can be created instantly while a call is happening. Right now also, some systems can already change faces and voices in real time with very small delay, which is enough to cheat video calls.

In the future, this will become even easier. With better software, faster chips, and mobile technology, high-quality deepfakes may work even on normal phones. This will make it possible for attackers to pretend as someone else during live video calls, meetings, or even customer support chats.

This can be used in many bad ways. For example, in online dating apps, money transfer apps, and social media, people can be tricked using fake identities. It can lead to fraud, identity theft, and even harassment.

The problem is, most current detection systems are made to check saved videos or images. But real-time deepfakes happen live, so it is much harder to detect. Because of this, better live detection tools will be needed in the future to handle these attacks.

3.2 Deepfake-Resistant Authentication and Verification Technologies

Because deepfake attacks are increasing, many researchers are now trying to build better security and verification systems. The main goal is to stop fake videos and fake identities from being used for fraud.

One idea is using systems that add a kind of digital proof when a photo or video is taken. This proof helps to check later whether the media is real or edited. Some standards like C2PA are working on this. Another idea is to build security directly into cameras and devices, so the video is verified from the start.

Also, new systems are being developed that use more than just face recognition. They combine things like how a person behaves, body signals, and other patterns to confirm identity. This makes it harder for deepfakes to trick the system.

In addition, some systems ask users to do random actions during verification, like blinking, turning head, or speaking a phrase. This makes it difficult for fake videos to pass the check, especially in bank KYC and similar systems.

3.3 Legislative and Regulatory Evolution

The laws about deepfakes will change a lot in the future. Right now, some countries like the US, Europe, China, and the UK already have some rules. These mainly deal with problems like fake private videos, election issues, and financial fraud. But the rules are still different in each country, and they are not fully connected.

In the future, stricter laws may come. People who create or spread harmful deepfakes may face stronger punishment. Also, AI-generated content may need to be clearly marked, especially in news, ads, and politics. This will help people understand what is real and what is not.

The European Union already started this with its AI law in 2024. It says AI content must be labeled clearly. This will affect media, advertising, and political content.

In the coming years, countries will need to work together more. Otherwise, people may misuse places with weak laws to spread deepfakes. So international cooperation will be very important.

3.4 AI-Powered Deepfake Detection at Scale

Right now, building strong AI systems to detect deepfakes is one of the most important areas in cybersecurity research. These systems can detect many known types of deepfakes quite well. But when new types of fake videos come out, the same systems often fail or give low accuracy. Attackers also take advantage of this by creating deepfakes that are designed to avoid detection.

In the future, researchers are trying to make better detection systems that can work in different situations, not just known cases. They are looking at using more than one type of information, like video, audio, and other signals together. They are also trying to find small signs in the body or behaviour that may show if something is fake. Training systems with large and varied fake data can also help improve accuracy.

Another idea is to add deepfake detection directly into apps like social media, video calls, and streaming platforms. This way, fake content can be detected while it is being shared. But this also brings challenges. Sometimes real content may get wrongly flagged, and it also needs a lot of computing power.

So the main challenge is to find a balance. The system should catch fake content properly, but at the same time, it should not disturb normal users or block real creative content. Clear rules and proper explanation to users will be important in the future.

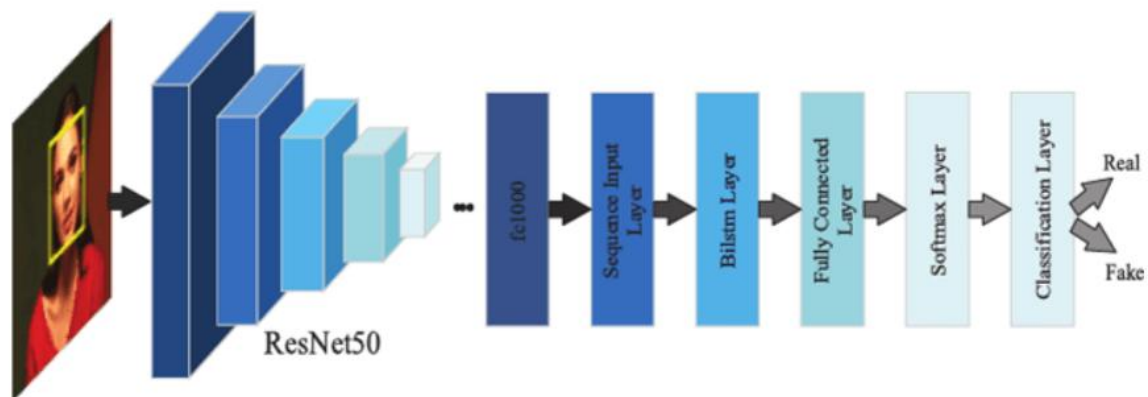


Figure 4: AI-Based Deepfake Detection Architecture using ResNet50 and BiLSTM, illustrating multi-layer classification into Real or Fake outputs

3.5 Quantum Computing Implications

Looking ahead, quantum computing might also affect deepfakes. If it becomes practical, it can make AI training much faster. That means future deepfakes could look even more real than today.

It may also become easier to create good quality fake videos because it won't need as much computing power. So more people could end up using it.

But quantum tech can also help in security. It can help build stronger ways to check and protect real videos and images. It can also help in sending and checking data more safely. Still, awareness is very important. Even if technology improves, people should know how to check what is real and what is fake. Tools alone are not enough.

So training and awareness should be given to everyone like students, workers, journalists, and normal users. Governments and organisations should help people understand this and stay safe.

3.6 The Role of Human Awareness and Social Impact

The real problem with deepfakes is not just the technology. It is also about how it messes with the way normal people think. When we watch a video, we just assume it is real. That is how we are wired. We trust what we see and hear. Deepfakes use exactly that against us. So even someone who is pretty careful can still get fooled because the fake looks way too convincing.

That is why people need to actually know about this. Tech alone is not going to fix it. People themselves have to understand what is going on and how to not fall for it. If someone already knows about deepfakes, they are going to think twice before trusting a random video or passing it on. They will check where it came from, notice if something looks a bit weird, and maybe confirm it through another way before doing anything about it.

Schools need to start teaching this. Students should know what deepfakes are, how they get made, and why it is a real issue. Kids who learn this stuff early are going to be way more prepared when they are older. And it is not just a tech thing. It is more about not blindly trusting everything you come across on the internet.

At the workplace, companies have to train their staff too. People in finance or management get targeted the most because attackers go after them with fake voice calls or fake video meetings. If workers know what to look for, they can avoid a lot of damage. Even something small like calling back on a separate number to double check a payment request can stop a scam right there.

Journalists have a big role in this as well. They push out news to thousands of people every single day. If someone in media shares a deepfake video without checking it, wrong info spreads like crazy. So people in journalism need to actually learn how to verify a video before

putting it out. News organisations should have proper tools for this and some kind of internal process to handle AI-generated content.

Social media is honestly where most of the damage happens. Fake videos go viral on these apps all the time. One deepfake can completely destroy someone's image or get people to believe something totally false about a well-known person. These platforms need faster and better ways to catch this stuff. At least putting a warning label on anything suspicious would help people think before they share it.

This whole thing hits society harder than most people realise. People are already starting to doubt videos and recordings they see online. That becomes a real problem in legal situations where video has always been treated as solid proof. If fake videos are this good now, how is anyone supposed to trust that evidence in court? Even judges and lawyers need to get familiar with this and use proper verification tools.

In politics it gets even worse. Fake videos of leaders are already being used to spread lies right before elections. People watch them and believe it. Some might even vote differently because of a video that was never real to begin with. That is a serious threat to how democracy works. Governments cannot just sit on this — they need to act before election season hits.

At the end of the day this is not just a tech problem to solve. Good tools help, sure. But you also need clear rules and people who actually get what is happening around them. When regular people, companies, governments and tech platforms are all on the same page, it gets a lot harder for attackers to pull this off. Nobody is saying stop the technology. The point is just that people should know enough to not get played by it.

4. Conclusion

The deepfake problem is global, so countries cannot handle it alone. It needs cooperation between governments, tech companies, and cybersecurity teams. Because everything online crosses borders, attacks also come from different countries. So sharing information, working together on investigations, and setting common rules are very important.

Deepfakes also affect society in a bigger way. The main issue is trust. People are slowly starting to doubt videos, audio, and images. This affects news, law, business decisions, and even democracy. When people cannot trust what they see online, it creates confusion. So along with technology, people also need better skills to check and understand digital content.

In the future, deepfake technology will become even more advanced. Real-time fake videos, better AI systems, and even faster computing like quantum computing can make deepfakes more powerful. At the same time, there are also solutions being developed like stronger security systems, better detection tools, improved face and voice checks, and new laws.

The problem is, deepfakes are growing faster than the tools used to stop them. So attackers are always one step ahead. There have already been real cases, like big financial frauds where companies lost millions of dollars, and voice cloning attacks on organisations. There are also services now where even unskilled people can buy deepfake tools and use them for attacks.

Overall, deepfake technology has changed very fast from a research idea in 2014 to a real cybersecurity threat today. Now it is used in fraud, fake news, scams, and even political influence.

In the future, research will focus on better detection systems, stronger laws that work across countries, and understanding how people react to fake media. Because as real and fake content become harder to separate, this will remain one of the biggest cybersecurity challenges in the coming years.

And it's not just big organisations that have to worry anymore. Like right now anybody can just go online, pay a little money and get a deepfake done. No coding, no technical stuff needed. Nothing. That's what makes it scary now because it used to be only hackers and tech people doing this. But now it's anyone. And the people getting hurt are not always some big company or politician. Half the time it's just a regular person who did absolutely nothing wrong. And those people have it the worst because they don't have lawyers or a whole team backing them up when something like this happens to them.

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